Lab Exercise 17- Creating Secrets in Kubernetes

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssword'
```

```
[mohdanas@Mohds-MacBook-Air Desktop % kubectl create namespace my-namespace
 namespace/my-namespace created
 mohdanas@Mohds-MacBook-Air Desktop % kubectl create secret generic my-secret \
   --from-literal=username=admin \
[  --from-literal=password='P@ssw0rd'
 secret/my-secret created
 mohdanas@Mohds-MacBook-Air Desktop %
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

The output will be base64 encoded. To decode, you can use:

```
echo -n “P@ssword” | base64
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

```
echo -n 'my-username' > ./username.txt



echo -n 'my-password' > ./password.txt
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl create secret generic my-file-secret --from-file=username
=./username.txt --from-file=password=./password.txt

secret/my-file-secret created
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
```

```
secret/my-registry-secret created
mohdanas@Mohds-MacBook-Air Desktop %
```

Verify the secret:

```
kubectl get secrets my-registry-secret -o yaml
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get secrets my-registry-secret -o yaml

apiVersion: v1
data:
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssword'
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f my-secret.yaml

secret/my-manifest-secret created
mohdanas@Mohds-MacBook-Air Desktop %
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

```
[mohdanas@Mohds-MacBook-Air Desktop % kubectl get secrets my-secret -o yaml
 apiVersion: v1
 data:
   password: UEBzc3cwcmQ=
   username: YWRtaW4=
 kind: Secret
 metadata:
   creationTimestamp: "2026-02-28T17:54:54Z"
   name: my-secret
   namespace: my-namespace
   resourceVersion: "12106"
   uid: f53f88ce-a9ca-44a2-90df-1f90ccedc96c
 type: Opaque
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
```

```
      key: username
  - name: PASSWORD
    valueFrom:
      secretKeyRef:
        name: my-secret
        key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f pod-with-secret.yaml

pod/pod-with-secret created
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASS
WORD'

USERNAME=admin
PASSWORD=P@ssw0rd
```

Step 7: Clean Up

Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret

secret "my-secret" deleted
secret "my-file-secret" deleted
secret "my-registry-secret" deleted
secret "my-manifest-secret" deleted
mohdanas@Mohds-MacBook-Air Desktop %
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted
```